

1.1 Gradient Descent vs Closed Form

L2 Norm:

$$\|w_{GD} - w_{CF}\|_2 = 4.0287 \times 10^{-9}$$

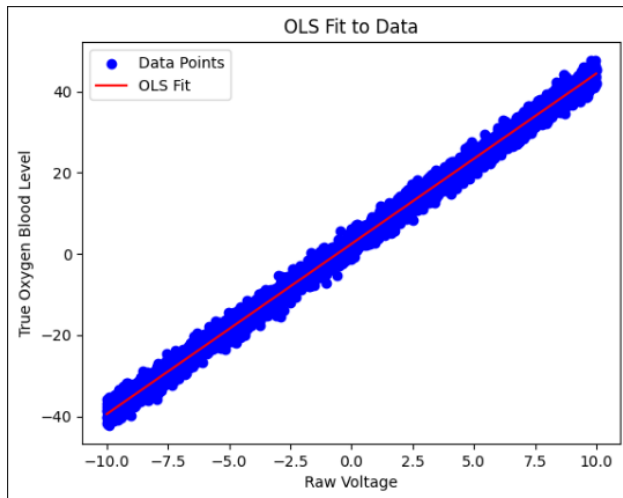


Figure 1: OLS fit for Dataset 1

1.2 Primal to Dual Translation

After thresholding the target variable at its median, a hard-margin SVM dual problem was formulated and optimized using `scipy.optimize.minimize`. However, the dataset is not linearly separable after thresholding, so the hard-margin constraints cannot be satisfied. This results in unstable optimization and indicates that a soft-margin SVM would be more appropriate.

2.3 The Singularity Trap

No issue as such was encountered while computing the OLS solution. However, the matrix condition number was calculated as $(\approx 2.25 \times 10^{19})$, indicating severe numerical instability. So the python exception may be referring to the singularity of the matrix, due to it having such a high condition number. Subsequently, L2 regularization was then used and requested graph plotted on the next page.

2.4 Lasso Sparsity

The generalized path plot was now drawn. Through an iterative search, the exact penalty required to drive exactly 50% of the feature weights (500 out of 1000 features) to zero was identified as $\lambda = 5.9806125123619385$

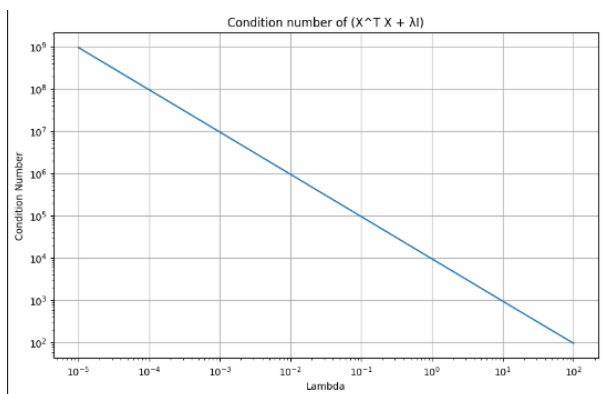


Figure 2: Condition Number vs. λ

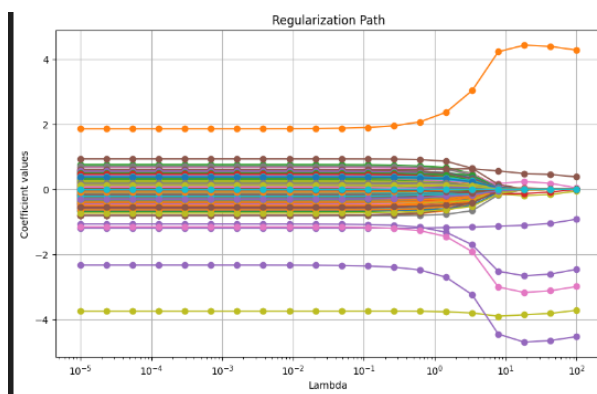


Figure 3: Coefficient Values vs. λ

3.5 Vectorized Logistic Regression

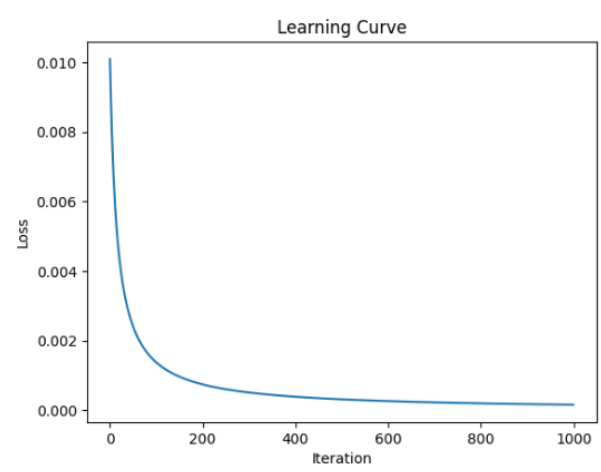


Figure 4: Learning Curve

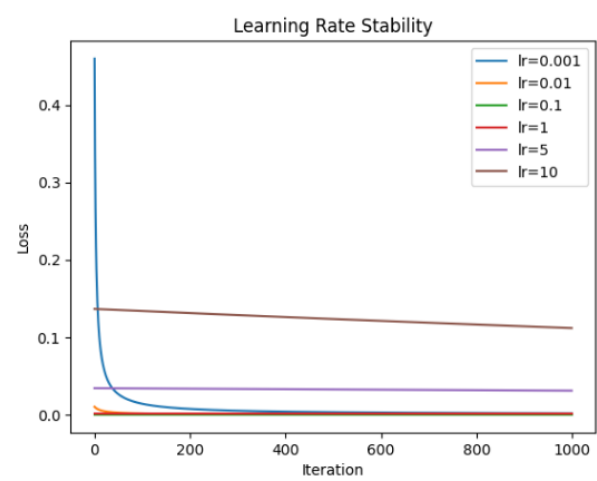


Figure 5: LR Lipschitz Violation

Lipschitz Smoothness Violation: Increasing the learning rate initially accelerates convergence, but pushing it too high causes the gradients to stagnate, to a point where they do not change at all even after 1000 iterations, thus depicting the Lipschitz Smoothness condition.

3.6 GMM Initialization & The E-Step

For $K = 4$, the initial marginal log-likelihood is -690775.52 , which improves sharply to -73815.04 after exactly one EM iteration.

3.7 EM Convergence & Crashes & 3.8 Decision Boundaries

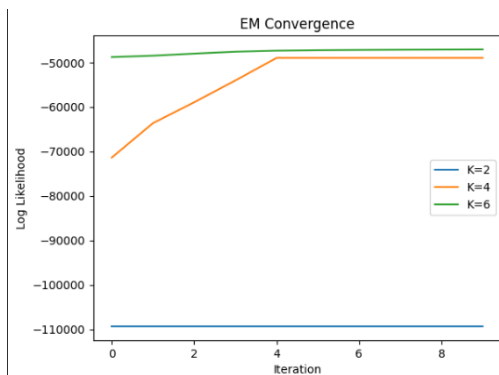


Figure 6: EM Convergence

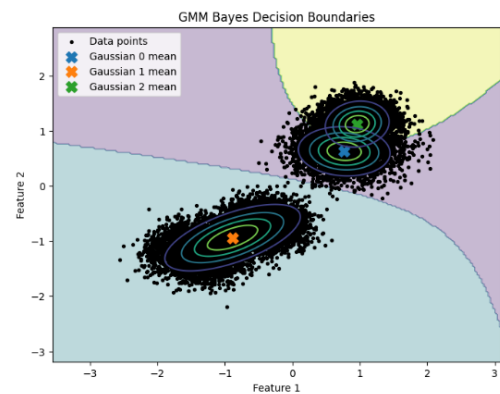


Figure 7: Bayes Decision Boundaries

Zero Variance Crash: Forcing zero variance mid-training breaks the multivariate normal PDF computation, yielding a `numpy.linalg.LinAlgError: Singular matrix exception`.

3.8 Decision Boundaries

Plot Interpretation: The curved, quadratic boundaries represent regions of maximum posterior probability $\pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$. Elliptical contours visualize the learned Gaussian covariances around the cluster means (marked by X).

3.9 SVM with Slack KKT Verification

KKT satisfied: True

Number of support vectors: 10

- (a) No point strictly inside the margin found.
- (b) On margin boundary: index = 30, $\mu = 0.001906$
- (c) Outside margin: index = 0, $\mu = 0.000000$

3.10 Mercer's Condition

Minimum eigenvalue: 0.028818695571655856

Maximum eigenvalue: 252.07943344080067

Since all eigenvalues are non-negative, the kernel matrix is positive semi-definite (PSD). Therefore, Mercer's condition is satisfied.

3.11 Hyperparameter Topography

As per the graph, we notice that the data has a very high accuracy regardless of the values of C and σ . However, as per the code, we see the worst overfitting is when $C = 0.1$ and $\gamma = 0.001$

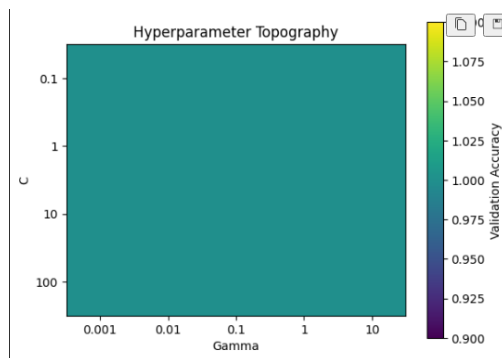


Figure 8: 2D Heatmap

4.12 Multiclass Extension & Scaling

In the scenario of Logistic Regression on Raw dataset, each class, using the one vs rest method, did not converge in 1000 iterations. The tolerance chosen was $1e - 3$, meaning that the 2 norm of the weight difference between two different iterations was still quite big. However, for the scaled data, in both cases of one vs rest (since there were 2 classes) the number of iterations taken to converge was 316 for both classes taken as one and the other as rest, of the scaled dataset.

4.13 Vectorization Benchmarking

Speedup Factor obtained upon using the vectorised version instead of the nested loop version was 94.41x.

4.14 The Curse of Scale

The KNN Accuracy on raw data was 0.9045, whereas on scaled data: 0.9210. Thus, for both cases of one vs rest, difference in KNN accuracy (scaled - raw): 0.0165

4.15 Bayes Optimal & Underflow

Under the given implementation, the accuracy using the Naive Bayes Classifier was 0.9133. However, no underflow was obtained at feature dimension, maybe due to the increased efficiency of computation via numpy. But the overall idea is the continued product method yields extremely small values, as probabilities upon being producted with each other result in values much lesser than 1, resulting in underflow.

5.16 Empirical Bias-Variance via Bootstrapping

- Linear Regression ($d = 1$)
 - Bias²: 2.3576982710523615
 - Variance: 0.0011163925183556942

- **Polynomial Regression ($d = 15$)**
 - Bias²: 154.87213749897555
 - Variance: 0.15907468752754955

5.17 Frequentist vs. Bayesian Uncertainty

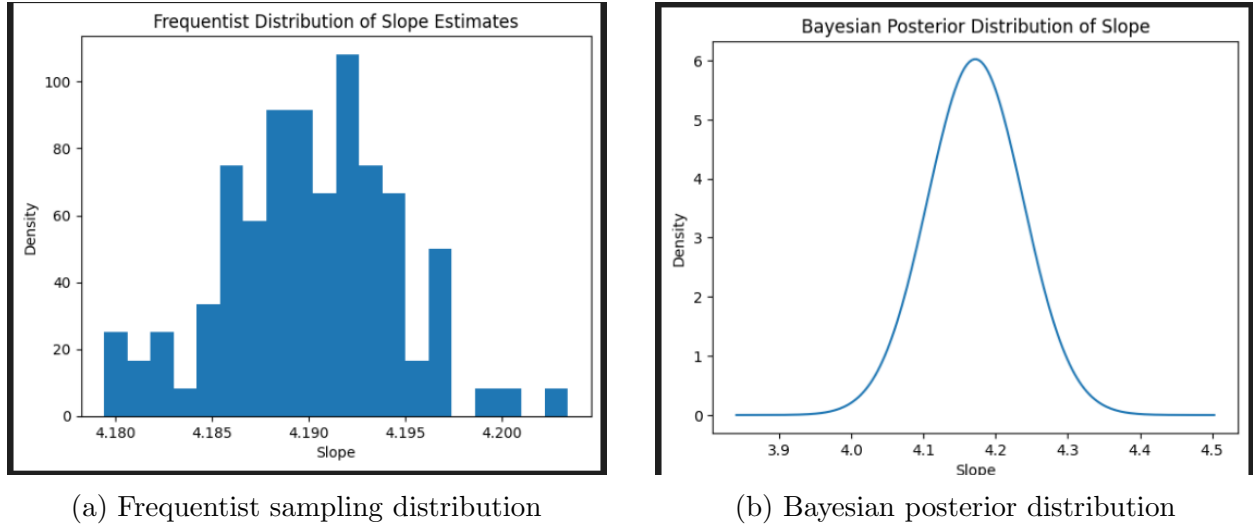


Figure 9: Frequentist vs Bayesian distributions of the slope parameter.

- **Frequentist Variance of slope parameter:** 0.000021
- **Bayesian Posterior Variance of slope parameter:** 0.004388

Interpretation:

- **Frequentist:** θ is fixed and the estimator $\hat{\theta}$ varies across sampled datasets, producing the bootstrap distribution.
- **Bayesian:** θ is treated as a random variable and uncertainty is represented by the posterior distribution $p(\theta | D)$.